McGill University

Honours Course Project

Supervisor: Prof. Humphries

Canards: a short life

for MATH 376: Honours Nonlinear Dynamics

Cedric Phillips

December 11th, 2023

1 Introduction

The first question that tends to spring to one's mind when they first encounter canards is probably "what is a canard?" or "a canard? Like a duck?" - these are reasonable questions, and yes, indeed, like a duck! Their name represents their shape when viewed in phase space. Much of their literature is very dense, and often is in foreign languages. This project aims to give an introductory and digestible overview of some of their properties, i.e., something that a student with only basic nonlinear dynamics experience could read and parse.

Canards can be found and analyzed in many ways. A little history lesson is required; they (in a published sense) first originate from the study of the *Van der Pol* oscillator, a very specific 2D relaxation oscillation where an incredibly miniscule parameter change causes the system to 'explode' from a smaller limit cycle into a large relaxation cycle. (Benoit et al.) We will cover the Van der Pol oscillator in more detail later, in order to motivate the rest of research.

Although variable parameters (say, μ), changes in dynamics based on these parameters, and limit cycles are familiar concepts from the lectures of MATH 376, relaxation cycles are perhaps unfamiliar. They are a special type of nonlinear limit cycle where at first there is a slow 'build-up' followed by a comparatively very sudden and quick 'release' or 'relaxation' (Strogatz 213). In fact the concept is not too disjointed from that of hysteresis seen earlier in the class. A diagram is illustrative:

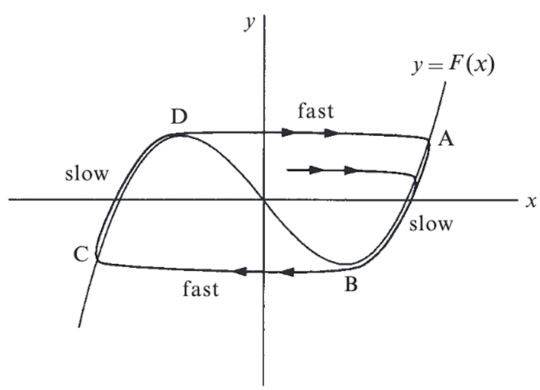

Figure 1: Relaxation cycles ($\mu \ll 1$) in the Van der Pol oscillator (Strogatz).

The crux is that our solutions change through a quick process, called *canard cycles*. They go from large amplitude stable trajectories to smaller amplitude unstable ones (in Figure 1, the solutions change from the form ABCD to a form ABD.) The language here is imprecise as we are attempting to gain a more intuitive and qualitative understanding of the situation(s) at hand; other mathematicians have already established the rigour.

2 The Van der Pol Oscillator

This particular system is an excellent gateway into the rest of the realm of canards. The Van der Pol oscillator is prototypically given as:

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$$

But through rearrangement (Lienard substitutiion $y = \varepsilon\dot{x} + \frac{x^3}{3} - x$ - see Zvonkin & Shubin 79) - we can generate a two-dimensional form aligning with a general form for canards which more clearly shows the results we want (note that we have $F(x) = \frac{x^3}{3} - x$):

$$\begin{aligned} \varepsilon\dot{x} &= y - F(x) = y - \frac{x^3}{3} + x \\ \dot{y} &= \mu - x \end{aligned} \tag{1}$$

With some positive real control parameter μ, and for ε small (say $\varepsilon = 0.01$). In this system we call y slow, since at all points on the plane, $\dot{y}$ has finite values. Conversely, x is fast as $\dot{x}$ has infinitely large values for some points on the plane. Then the slow curve is defined as the points where $\dot{x}$ vanishes, i.e $\dot{x} = 0$, or $F(x) = \frac{x^3}{3} - x$. Then y attains a local minimum at $x = 1$ (and of course a maximum at $x = -1$). The only fixed point is $(\mu, F(\mu))$. Much like in the content of 376, we linearize at the fixed point:

$$J = \begin{pmatrix} -\frac{1}{\varepsilon}F'(\mu) & \frac{1}{\varepsilon} \\ -1 & 0 \end{pmatrix} \text{ so } J(\mu, F(\mu)) = \begin{pmatrix} -\frac{1}{\varepsilon}F'(\mu) & \frac{1}{\varepsilon} \\ -1 & 0 \end{pmatrix}$$

$$\text{Then we find the eigenvalues: } \lambda = \frac{-F'(\mu) \pm \sqrt{(F'(\mu))^2 - 4\varepsilon}}{2\varepsilon}$$

We then have that the fixed point is stable for $F'(\mu) > 2\sqrt{\varepsilon}$, but since we take very small ε, we effectively have stability for $F'(\mu) > 0$. Conversely, it is unstable for $F'(\mu) < 2\sqrt{\varepsilon}$ or rather $F'(\mu) < 0$. There is a little more nuance here vis-a-vis the small epsilons but it is not particularly useful for the overview we are giving. Notably, linearization fails for $F'(\mu) = 0$ (i.e at $|x| = 1$) as both eigenvalues are imaginary. There is a lengthy computation (see Zvonkin) that delivers for us the result that at $|x| = 1$, the fixed point is a stable focus.

Much like we have done in class, we must consider the system case-by-case for different values of μ; for simplicity, we focus mainly on the positive-x side of the situation. For $\mu \ll 1$, there exists only one large relaxation cycle. We characterize the portions of $F(x) = y$ where $x < -1$ and $x > 1$ as increasing and furthermore attracting, and the portion $-1 < x < 1$ as decreasing and repulsive. This is logical since we determined that the fixed point is stable (i.e attracting) for $F'(\mu) > 0$, and vice-versa for when $F'(\mu) < 0$. Then since we have $\mu \ll 1$, the center "decreasing" portion is where the unstable fixed point is located, and so since solutions do not eventually end up at this point, we obtain the relaxation cycles.

Crucially, the cycles never overlap as they never truly reach $F(x)$; they begin far away, move parallel to the x axis towards $F(x)$, come (infinitely) close to it (staying in its "halo"). Once in its halo, a solution will slowly follow $F(x)$ until it reaches a "summit", at which point it will transfer to a horizontal "fast" course until it again reaches $F(x)$. It is this transfer to the horizontal course where canards appear later, but we are safe from them for now. This process is seen in Figure 2.

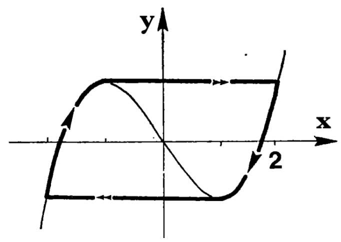

Figure 2: The stable relaxation cycle, shown with amplitude; $\mu < 0.998$ (Benoit et al.).

Furthermore, this relaxation cycle has maximum amplitude of 4, is unique and is asymptotically stable. The amplitude derives from the fact that $F(x)$'s value at its summits is $|\frac{2}{3}|$; then $F(x)$ only reaches this again at $|x| = 2$. The rigorous proof of uniqueness and stability are not given here (they are very complex) but can be found in (Benoit et al., 49).

Our next case is if $\mu > 1$. Here there are no cycles. This is because now $F'(\mu) > 0$, so the fixed point is stable, and furthermore since the fixed point has coordinates $(\mu, F(\mu))$, it is on the slow part of the would-be relaxation cycles. In this instance, although a solution may begin as a cycle, passing through the usual steps and jumps, it will eventually encounter the fixed point and stop there. This is the sole fixed point of the system.

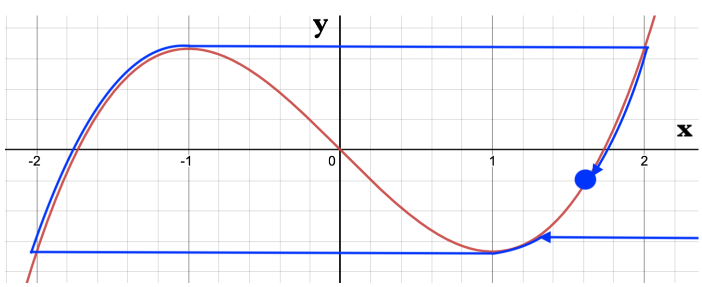

Figure 3: Development of a solution towards stable FP for $\mu > 1$ (Desmos).

2.1 Canard

So clearly there is some strange process occurring between smaller values of μ and ones greater than 1. Without spoiling too much for the moment, part of the mess is that there is a Hopf bifurcation at $\mu = 1$. But we will get to that later - it is distinct from the focus of our studies. We are closing in on a canard. We know there is a relaxation cycle for $\mu \ll 1$, but what happens when $\mu \sim 1$, i.e, μ is *almost* 1?

It seems that previously, for $\mu \ll 1$, a solution following $F(x)$ would jump off once it reached a summit, and thereafter continue on the other side of $F(x)$. But for μ close enough to a summit (i.e at 1), this reliable phenomenon seems to break down, leading to canards. In a simplified way, this is effectively because the cyclic solution is never truly on $F(x)$, staying in its halo; when the value of μ is so close to 1 that μ is effectively within the halo of the summit, solutions can end up continuing along the decreasing part of $F(x)$. These solutions are still cyclic, so we call them *canard cycles*.

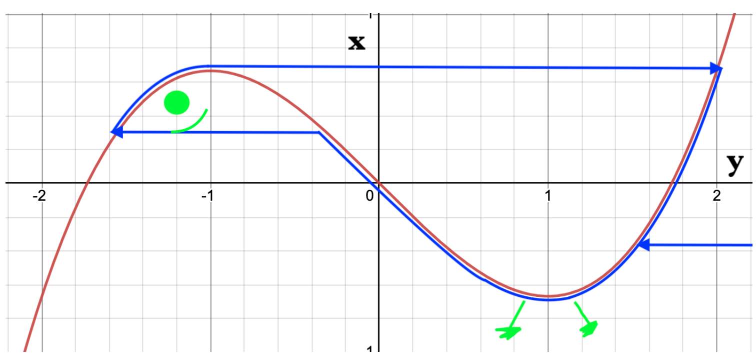

Figure 4: Diagram of a prototypical canard cycle. Marks in light green have no mathematical bearing, and simply demonstrate nomenclature (Desmos).

Then it is possible that the solutions jump either to the left onto the negative-x increasing part of $F(x)$, wherein we say that the canard has a bill, or to the right. This is not random; as μ increases the canard's bill will disappear; past a certain value of μ, every solution jumps in the positive-x direction, and the canard cycles quickly decrease in amplitude, moving towards the fixed point $x = 1$ as μ also approaches it.

It can be seen (Figure 5) that once μ has exceeded a value of about 0.998, the relaxation cycles end and the canard cycles begin, and in very short order, the amplitude of the cycle is quickly reduced to near zero. A 2×10^{-12} change in μ causes the amplitude to drop from about 3.5 to about 0.1, and it trends to 0 with any further increase in μ .

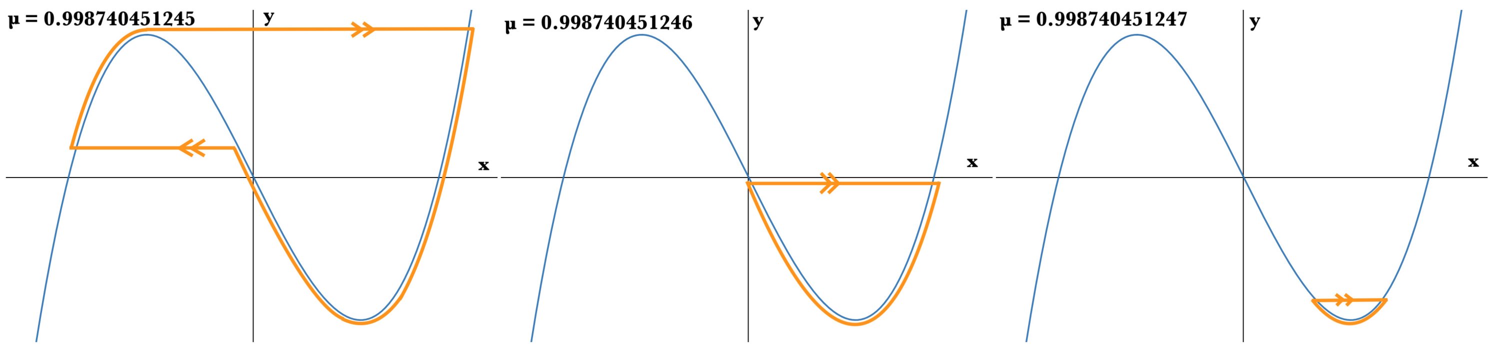

Figure 5: Diagram of canard cycle for very small changes in μ. Note: in the third diagram, the size of the cycle is exaggerated so we can see it; in reality, it is already minuscule.

An interesting remark on the canard cycles here is that although they are highly influenced by changes in the parameter, in the Van der Pol oscillator, they are actually asymptotically stable for a constant parameter value. This follows from evaluating a certain integral and divergence (Zvonkin & Shubin 119). Since every canard is stable, this also implies that for every value of μ, there is only one canard cycle, as if there were more they would have to all be stable - but as we know from the lectures of 376, there cannot be two neighbouring solutions that are both stable.

2.2 Hopf Bifurcation

We now know that the canard cycles rapidly decrease as μ increases. Still, that does not exactly explain the transition from stable cycles (for $\mu < 1$) to a stable fixed point (for $\mu > 1$). Something is up. We have had these cycles around this point for quite some time, and now the cycles disappear, leaving only a stable point. Intuitively this is a Hopf bifurcation, and moreover, it is supercritical; on one side of it is an unstable fixed point surrounded by a stable limit cycle, and on the other side a stable attractor. The behaviour is logical; as μ increases, the canard cycles become infinitely small until they coalesce into a single point at $\mu = 1$.

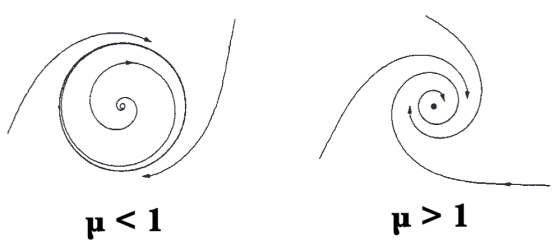

Figure 6: Supercritical Hopf bifurcation near $\mu = 1$ (adapted, Benoit et al.)

We are nearly done with the Van der Pol oscillator; in a concluding remark, we note that the range of μ where the canard cycles occur is not seemingly random and is indeed tied to the expression $e^{-\frac{1}{k\varepsilon}}$, for some positive finite k. In our investigation, we had $\varepsilon = 0.01$, so the range of canards for μ is at least about $e^{-10^5} \approx 10^{-44}$; (Zvonkin & Shubin 108) this is what we mean when we say the life of canards is short.

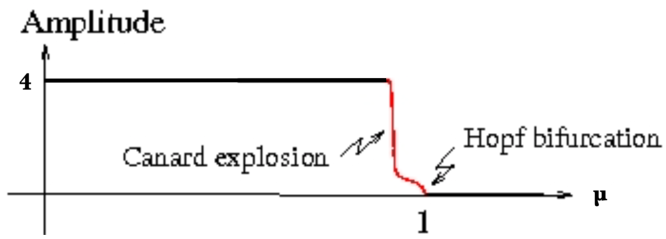

Figure 7: Overview of relaxation cycles, subsequent canard "explosion" and Hopf bifurcation (adapted from Wechselberger).

3 Generalization

We can now consider a more general meaning of a canard system; it is a slow-fast dynamic system where there seems to be a "missing middle" because of the very quick change in amplitude. Moreover, canards are not discontinuities. There exist many other canards than the one found in the Van der Pol oscillator. We focused on that one system because compared to many others, it is relatively easy to grasp and provides a more intuitive approach to the hunt for other canards. Moreover, many of the other systems which produce canards are of a similar form as Van der Pol's system.

3.1 "Cubic" or S-Shaped Systems

Consider a system similar to (1), but without defining $F(x)$ explicitly:

$$\begin{aligned} \varepsilon\dot{x} &= y - F(x) \\ \dot{y} &= \mu - x \end{aligned} \tag{2}$$

Such a system will have canard cycles in the interval $[a, b]$ if $F(x)$ is a standard function with continuous second derivatives on $[a, b]$, and if $F(x)$ has isolated minimum at x_0 and maximum at x_1 on $[a, b]$. For simplicity we take $a < x_1 < x_0 < b$. Moreover, we must have $F(a) < f(x_0)$ and $F(b) < f(x_1)$, i.e, the absolute minimum and maximum of the interval are at a and b respectively (Zvonkin & Shubin 114).

3.2 Slow-Fast Systems

In that sense the Van der Pol oscillator is a very simple example of a system with "cubic" behaviour that yields canard cycles. This of course does not mean that canards only exist in these sorts of systems; they can exist in many other systems that are slow-fast. We write these generally:

$$\begin{aligned} \varepsilon \dot{x} &= f(x, y, \varepsilon, \mu) \\ \dot{y} &= g(x, y, \varepsilon, \mu) \end{aligned} \tag{3}$$

With the same conventions as before. Canards can exist in systems of dimension greater than 2, i.e, x, y are variables in $\mathbb{R}^n$. Systems in $\mathbb{R}^3$ are a little trickier, but can still be visualized. Higher dimensions are more difficult to analyze, though they can sometimes be simplified into a two or three-dimensional system (Wechselberger). The behaviour of these systems is, by definition, similar to that of the Van der Pol oscillator; the canards are the solutions which after following the attracting slow curve for a while, will refuse, at some critical point, to migrate via the fast curve and will instead follow the slow curve further for some time (Diener 40).

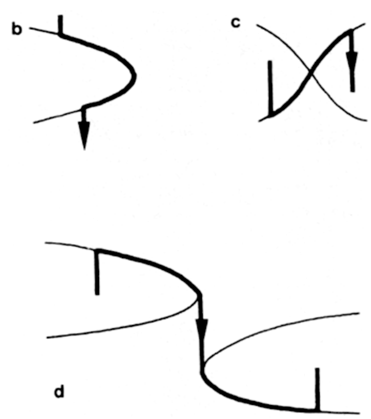

Figure 8: Canard varieties: in a), a familiar one, in b), a canard at a Morse-point, and in c), a canard which jumps to the critical point of another curve (Diener).

Another property of canards that we saw exemplified in the Van der Pol oscillator is that they are associated with nearby bifurcations. This is logical as they are the "middle" stages between whatever is occurring on the two sides of the bifurcation. These bifurcations are often Hopf ones but are not necessarily so; for example, the system in c) of Figure 8 is associated with what is called a "Heaven-and-Hell" bifurcation (Diener). Suffice to say, the systems in which canards arise are many, diverse, and they are not always cut-and-dry. For this reason we employ diverse techniques.

4 The Blow-Up Method

All this discussion and dissection has been done through the methods of what is now generally called "non-standard analysis". Although these methods may not indeed be standard, they are more intuitive, for beginners. Now we discuss one of the most common standard methods: blow-up. Although we just mentioned that canards can be found in non S-shaped systems, we are introducing a complex process and therefore will be working with S-shaped systems and Hopf bifurcations. The techniques laid out here, however, can be applied to other systems as well (Krupa & Szmolyan (a)).

4.1 The Canonical Form

Since we are now more generalized, we begin by making a shift from the previous Van der Pol system, such that the "canard point" is now around $(x, y) = (0, 0)$, which is the minimum of our manifold function f and is also where $\mu = 0$. The sort of equation in (3) is typically rewritten in x', y' for the purposes of blow-up, and we show the transformation here in order to align with the literature (Krupa & Szmolyan (b), 322).

$$\begin{aligned} x' &= -yh_1(x, y, \varepsilon, \mu) + x^2 h_2(x, y, \varepsilon, \mu) + \varepsilon h_3(x, y, \varepsilon, \mu) \\ y' &= \varepsilon(xh_4(x, y, \varepsilon, \mu) - \mu h_5(x, y, \varepsilon, \mu) + yh_6(x, y, \varepsilon, \mu)) \end{aligned} \tag{4}$$

Where $h_j(x, y, \varepsilon, \mu) = 1 + O(x, y, \varepsilon, \mu), j \neq 3$, and $h_3(x, y, \varepsilon, \mu) = O(x, y, \varepsilon, \mu)$, with O being (as standard) the order/growth rate of the function.

4.2 The Blow-Up Map

When we "blow up" a system, we define a map that essentially allows us to "zoom in". It is a difficult and computationally heavy process, but the analysis can be more nuanced in a blown-up view. Let S be a manifold such that $S = \{(x, y) : y = f(x)\}$. Define $\varepsilon_0 > 0$ such that $\varepsilon \in (0, \varepsilon_0]$. Then the blow-up transformation Φ is given as a mapping:

$$\Phi : B = S^2 \times [-\lambda, \lambda] \times [0, \rho] \to \mathbb{R}^4 \tag{5}$$

Where S^2 is a sphere, $\rho > 0$ is given by $\varepsilon_0 = \rho^2$, $\lambda > 0$ is given by $\lambda \leq \rho$, and the details of the mapping follow:

$$x = \overline{rx}, \quad y = \overline{r}^2\overline{y}, \quad \varepsilon = \overline{r}^2\overline{\varepsilon}, \quad \mu = \overline{r\mu} \tag{6}$$

With the powers of r acting as weights, and $(\overline{x}, \overline{y}, \overline{\varepsilon}, \overline{\mu}) \in S^2$. The details of this are hard to digest, and there is no real simple way to put them. We can say that at least what we truly want from ρ and λ is that they are small.

We use charts K_1 and K_2 to analyze the blown-up vector field of a system. K_1 is obtained with $\overline{y} = 1$ and K_2 is obtained for $\overline{\varepsilon} = 1$. There also exist K_3 and K_4, but K_2 (and to an extent K_1) is the most illustrative. Then the following transforms are defined:

$$\begin{aligned} \Phi_{K_1} &: x = r_1 x_1, \quad y = r_1^2, \quad \varepsilon = r_2^2 \varepsilon_2, \quad \mu = r_1 \mu_1 \\ \Phi_{K_2} &: x = r_2 x_2, \quad y = r_2^2 y_2, \quad \varepsilon = r_2^2, \quad \mu = r_2 \mu_2 \end{aligned} \tag{7}$$

With $(x_1, y_1, \varepsilon_1, \mu_1), (x_2, y_2, \varepsilon_2, \mu_2) \in \mathbb{R}^4$. In a typical analysis of a system using this blow-up technique, one now would redefine essentially everything we covered in sections 2 and 3 in the context of the blown-up system. This process is arduous and not very space-efficient; it is detailed in (Krupa & Szmolyan (b)). We skip ahead a little, but need to define a few more entities:

$$a_1 = \frac{\partial h_3}{\partial x}(0,0,0,0), \quad a_2 = \frac{\partial h_1}{\partial x}(0,0,0,0), \quad a_3 = \frac{\partial h_2}{\partial x}(0,0,0,0)$$

$$a_4 = \frac{\partial h_4}{\partial x}(0,0,0,0), \quad a_5 = h_6(0,0,0,0)$$

We also define the constant A as follows:

$$A = -a_2 + 3a_3 - 2a_4 - 2a_5 \tag{8}$$

A is useful as its sign is indicative of the dynamics related to canard cycles. The author understands that without the full context, many of these values seem to be plucked out of thin air. There is solid mathematical motivation behind all of them; it is just quite complex, and the goal of this project is not to unnecessarily further complicate the matter at hand.

We now want to generalize some "curves of μ". Suppose we have an S-shaped curve that will contain canards. Then for $\varepsilon \in (0, \varepsilon_0]$, for $|\mu|$ less than some certain μ_0, there is always an equilibrium p converging to the canard point when $\mu \to 0$. There is also a curve μ_H such that p is stable for $\mu < \mu_H$, and that there is a Hopf bifurcation as μ passes μ_H. We give μ_H:

$$\mu_H(\sqrt{\varepsilon}) = -\varepsilon \frac{a_1 + a_5}{2} + O(\varepsilon^{\frac{3}{2}})$$

This Hopf bifurcation is supercritical for $A < 0$ (like in the Van der Pol equation) and subcritical for $A > 0$. At $A = 0$ it is degenerate. We also define μ related to a function μ_c (read: mu canard) such that the slow parts of the system will only connect if $\mu = \mu_c$:

$$\mu_c(\sqrt{\varepsilon}) = -\left(\frac{a_1 + a_5}{2} + \frac{A}{8}\right)\varepsilon + O(\varepsilon^{\frac{3}{2}})$$

Finally, let $v \in (0,1)$ and write $\mu_s(\sqrt{\varepsilon}) = \mu(\varepsilon^v, \sqrt{\varepsilon}), \mu_r(\sqrt{\varepsilon}) = \mu(2y_{MAX} - \varepsilon^v, \sqrt{\varepsilon})$ (Krupa & Szmolyan (b) 327). These two mu mark the ends of the canard cycles, with s standing for the "small" cycle and r for the relaxation one. Then we have:

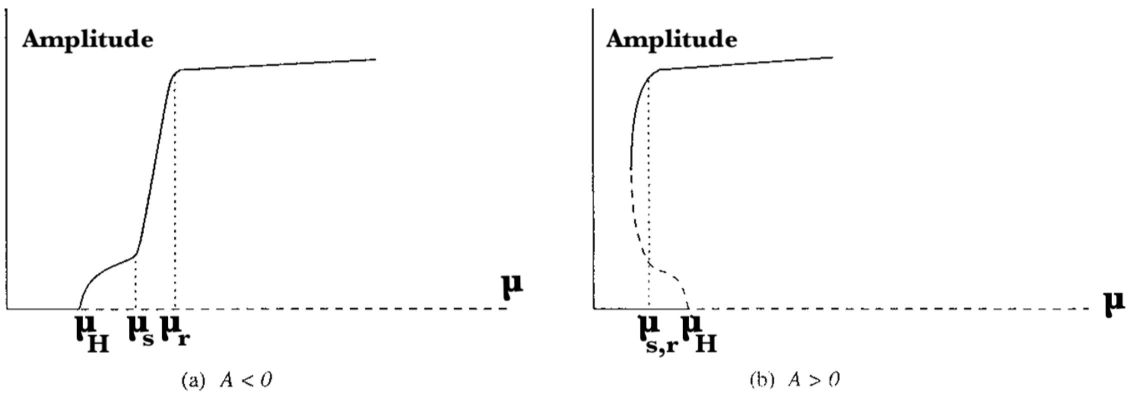

Figure 9: Dynamics on μ-curves for general S-shaped systems, by result of blow-up transformation, with the "short life" of a canard between the μ-curves. Hopf bifurcation as A changes sign (adapted, Krupa & Szmolyan (b)).

We would also like to show how blow-up lets us categorize dynamics in ways we could not before. If we take a look at the system transformed by (4)'s chart of K_2 for different "μ areas", we can see how the dynamics vary. Let $\mu_c(\sqrt{\varepsilon}) - \mu_{sc}(\sqrt{\varepsilon}) = O(e^{-\frac{1}{k\varepsilon}})$, which, if one recalls section 2.2, is the generalized equivalent of the parameter range for which canard cycles are in progress. Then the following diagram may seem intuitive:

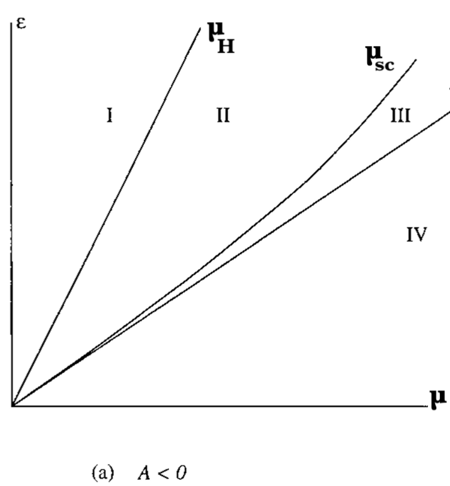

Figure 10: "μ areas"; regions of typical behaviour between supercritical Hopf bifurcation curves around canard point (adapted, Krupa & Szmolyan (b)).

Finally we can see how the system behaves for each region, in the phase portrait of K_2 in Figure 11. Note that $S_{l,\varepsilon}$ and $S_{m,\varepsilon}$ are slow manifolds; one can analogue with the Van der Pol Oscillator that $S_{l,\varepsilon}$ is the "outside" part and that $S_{m,\varepsilon}$ is the "inside" between the minimum and maximum.

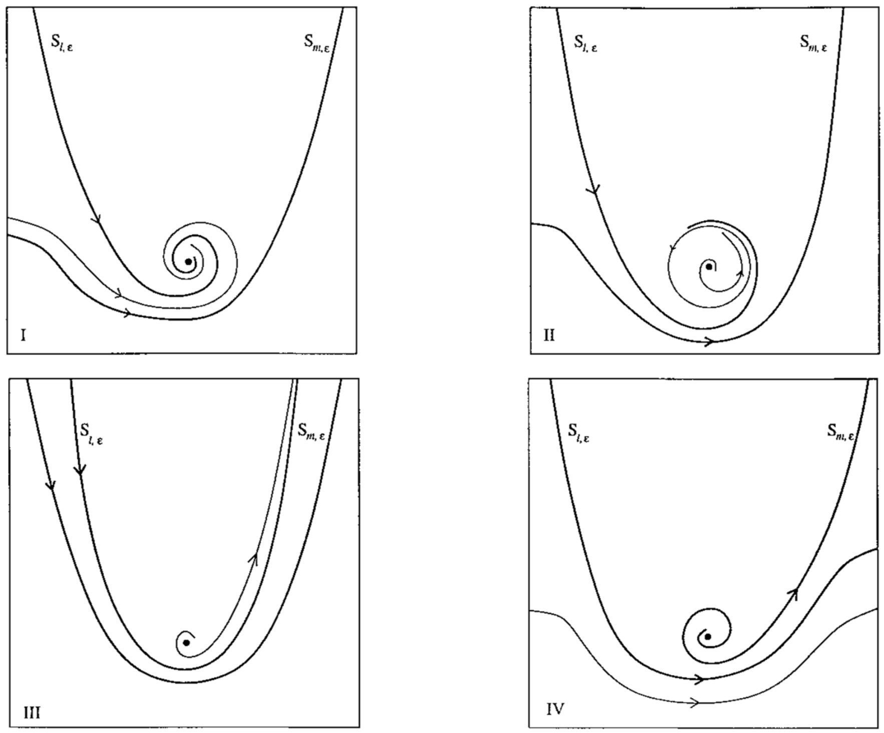

Figure 11: The general behaviour in the K_2 chart, where a supercritical Hopf bifurcation can clearly be seen ($A < 0$), as well as a glimpse of canard cycles, as we pass through the various "μ areas" (Krupa & Szmolyan (b)).

This blow-up analysis allows one to get more "face-to-face" with the dynamics for S-shaped canard systems, including those with subcritical Hopf bifurcations, by appropriate transformation; one does not have to start from scratch with each system. From here one can continue to use the K charts to analyze the behaviour of systems of one's choice. There are more in-depth ways to investigate canards once the blow-up transformation has been applied, but they are very dense and do not necessarily yield any spectacular results. Perhaps one of the ways in which blow-up is most useful is how it can quickly categorize the Hopf bifurcation in S-shaped systems. The existence of the constant A makes this a simple calculation which could be easily computed. Since the Hopf bifurcation informs on the canard cycles to an extent, it is a great place to start; if one didn't want to continue the canard analysis in blown-up space, they could return to traditional coordinates after better understanding the system's behaviour near the canard point.

5 Other Applications and Conclusion

5.1 Predator-Prey Systems

We have studied predator-prey systems in MATH 376, and they, too, given a damping effect, can show canards. In a slow-fast scenario where predators very rarely die, it was seen (via blow-up methods) that when the prey had access to a group defense mechanism, there was a steady relaxation cycle that at some point devolved down into a single point via canards (Li et al.). Yet this paper is more interesting in the other direction, as a growth via canard cycles could explain why some long nearly extinct animal species seem to sometimes miraculously explode in population and back into the ecosystem.

5.2 Chemistry

The author of this research project has dabbled in the field of chemistry to an extent (not a great extent, but an extent nonetheless) and is familiar with the iodine clock reaction. In this reaction, two colourless solutions are mixed, and thereafter stay colourless for quite some time, slowly mixing with each other but producing no discernible change. Then, suddenly, the entire mixture turns dark purple in the blink of an eye, and stays that way. In an analogy to canards, we have a stable (perhaps relaxation) cycle until a miniscule change in the conditions causes the system we are dealing with to rapidly change (via canard cycles or the process of a chemical reaction) to a different sort of steady state. Whether or not the dynamics of the iodine clock reactions truly contain canards in a mathematical sense is not covered here, but perhaps it is in intriguing avenue for future research. Regardless, we deem the iodine clock systems to be in the same spirit as the systems we here have investigated. In general, reactions that have "hard transitions" are likely to involve canards, as one cannot have a discontinuous reaction.

5.3 Conclusion

Although we only barely began to scratch the surface of the possibilities of blow-up, there is only so much that can be covered in project like this that is meant to be "accessible". We took a detailed look at the Van der Pol oscillator in order to understand the most basic concepts of canards, and then covered one of the modern techniques used to analyze these types of systems. Much of the literature on this subject is extremely recent, and, in terms of problems and systems unexplored, the iron is still hot. Hopefully we have elucidated some of the concepts herein to lower-undergraduate reader, and given them perhaps some inspiration to seek out canards, or at least provided them with some tools if they do encounter them.

6 References

1. Benoit, E., Diener, F., & Marc Diener. (1981). Chasse au canard. *Collectanea Mathematica*, 32(1).

2. Diener, M. (1984). The canard unchained or how fast/slow dynamical systems bifurcate. *The Mathematical Intelligencer*, 6(3), 38–49. doi:10.1007/bf03024127

3. Graphing calculator. (n.d.). *Desmos*. https://www.desmos.com/calculator

4. Krupa, M., & Szmolyan, P. (2001, (b)). Relaxation oscillation and canard explosion. *Journal of Differential Equations*, 174(2), 312–368. doi:10.1006/jdeq.2000.3929

5. Krupa, M., & Szmolyan, P. (2001, (a)). Extending geometric singular perturbation theory to nonhyperbolic points. *SIAM Journal on Mathematical Analysis*, 33(2), 286–314. doi:10.1137/s0036141099360919

6. Li, Q., Zhang, Y., & Xiao, Y. (2023). Canard phenomena for a slow-fast predator-prey system with group defense of the prey. *Journal of Mathematical Analysis and Applications*, 527(1), 127418. doi:10.1016/j.jmaa.2023.127418

7. Strogatz, S. (2015). *Nonlinear Dynamics and Chaos*. S.l.: CHAPMAN & HALL CRC.

8. Wechselberger, M. (2007). Canards. *Scholarpedia*. http://www.scholarpedia.org/article/Canards

9. Zvonkin, A., & Shubin, M. (1984). Non-standard analysis and singular perturbations of ordinary differential equations. *Russian Mathematical Surveys*, 39(2), 69–131. doi:10.1070/rm1984v039n02abeh003091